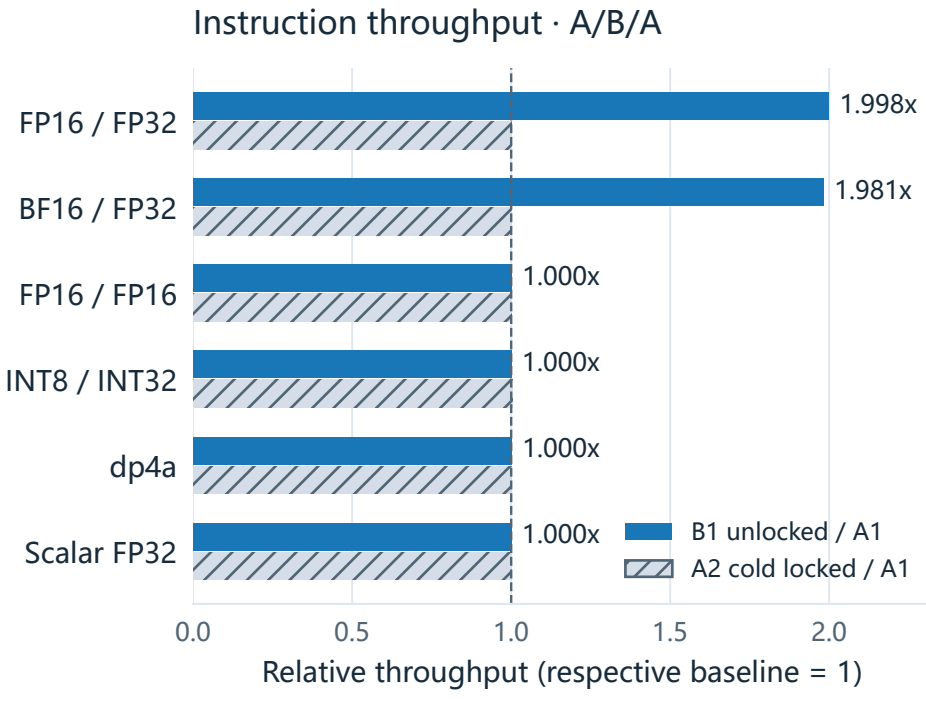
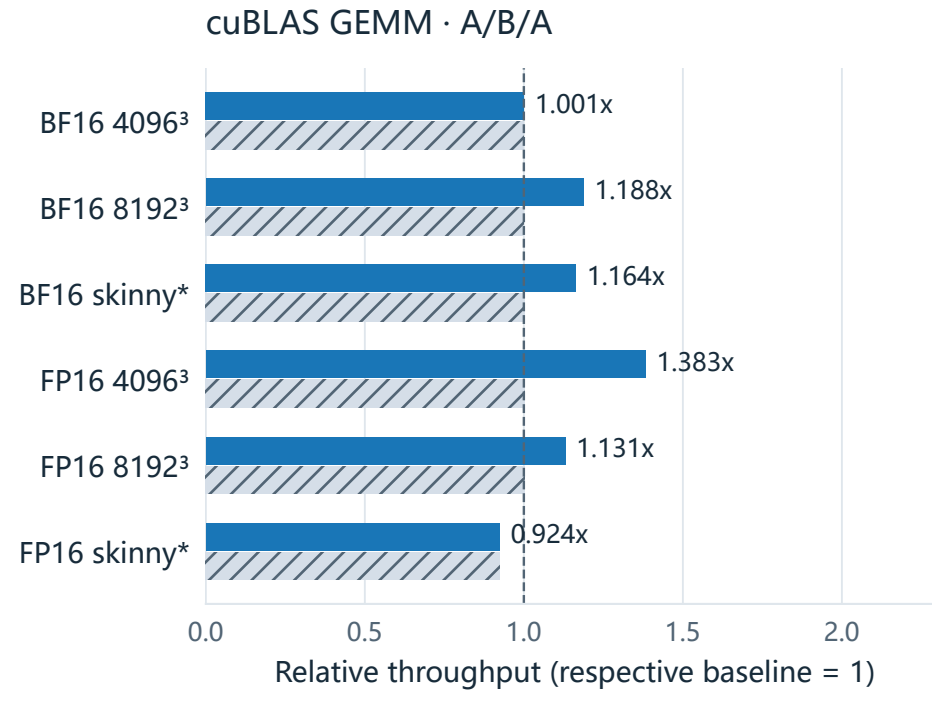
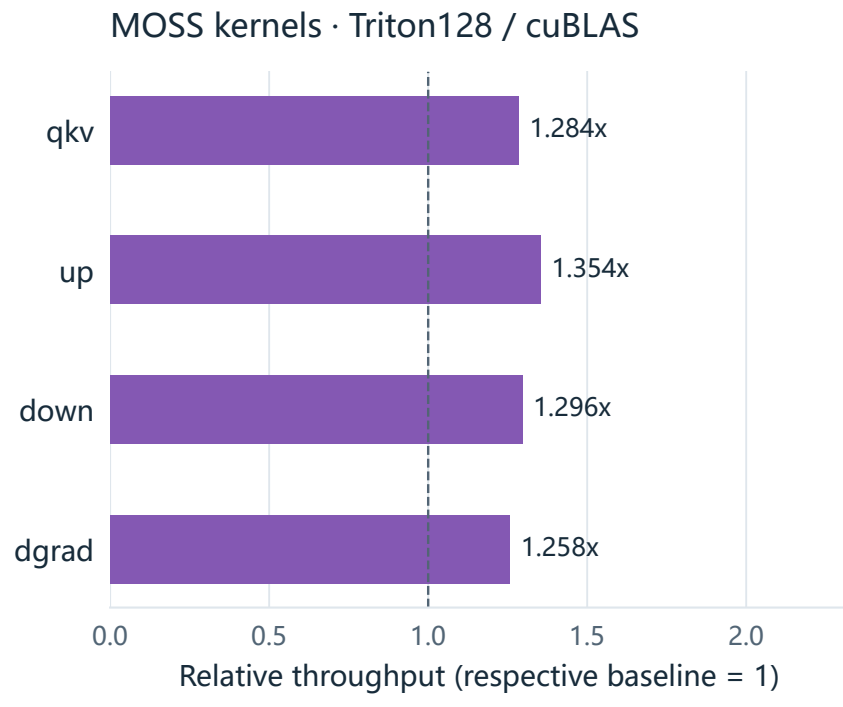
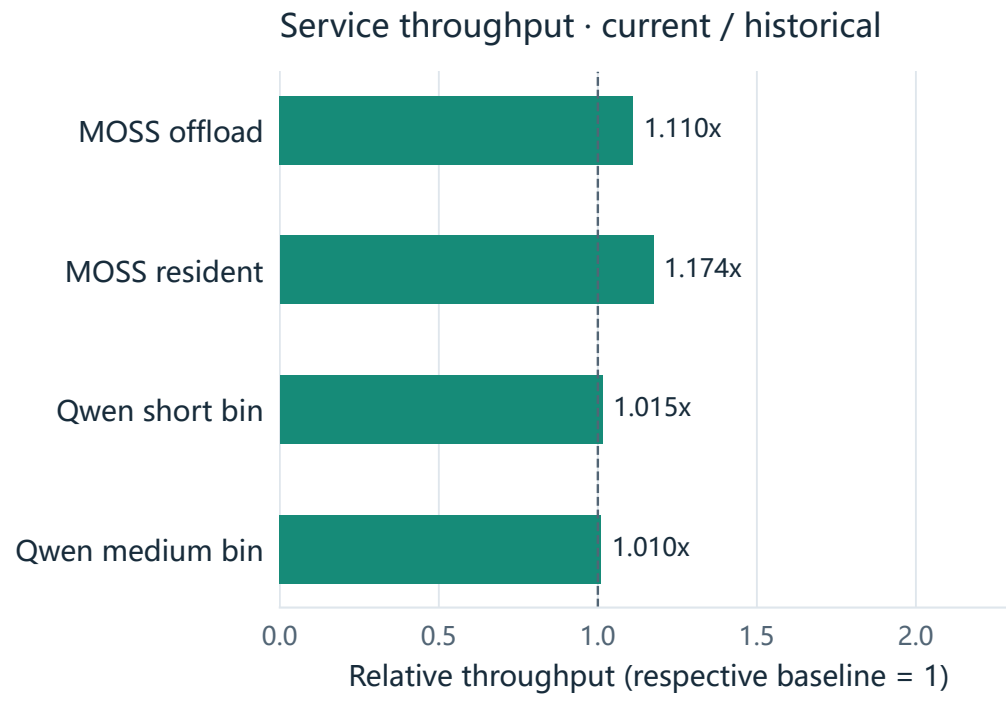
Performance across instructions, kernels, and services

Dual RTX 4090 · 48 GiB/card · 450 W/card









* skinny = 256x512x16384; instruction n=30/mode, GEMM n=10/shape.

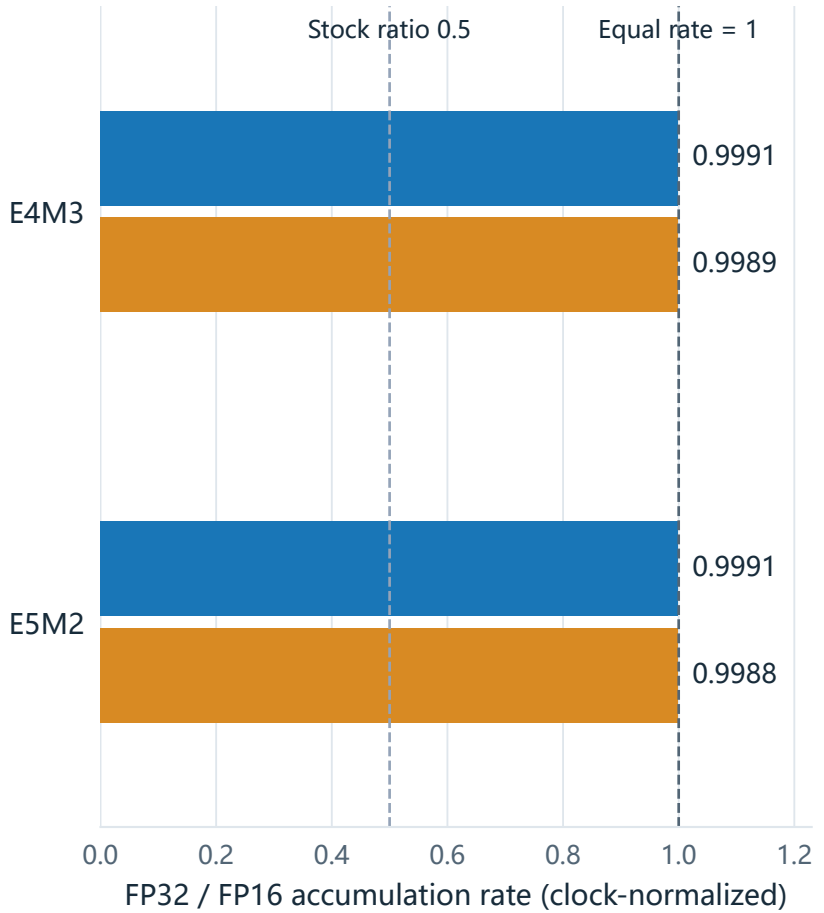
Sources: ab-a-matched.json | operators.json | historical-applications.json (2026-10-26)

Native FP8 throughput

Both cards unlocked · n=21/mode · dense operations · whiskers: min-max

Stock specification and current measurements

FP32 versus FP16 accumulation on each card

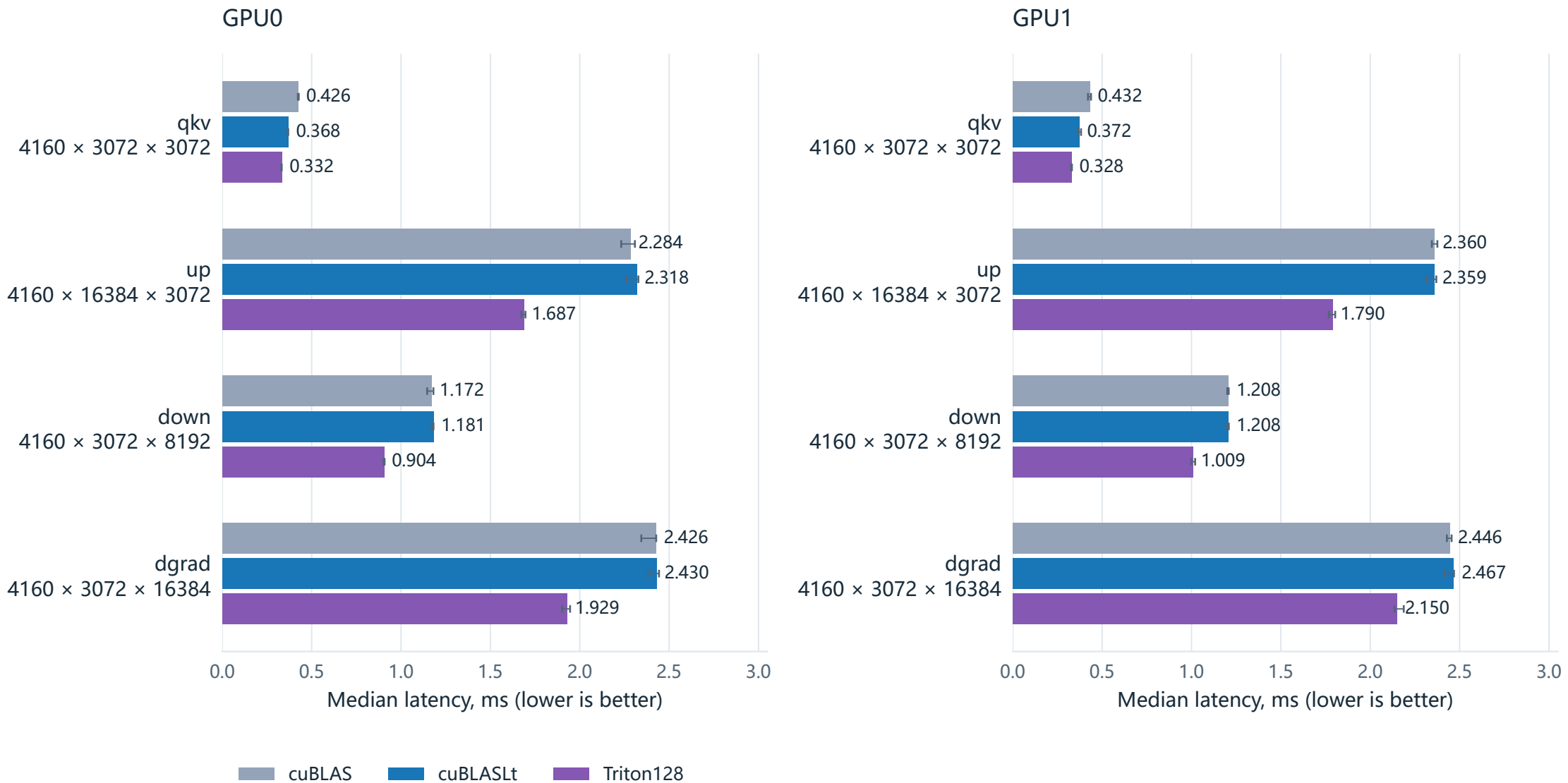


Clocks: GPU0 ~2710 MHz · GPU1 ~2744 MHz · stock specification 2520 MHz.

Source: 2026-10-29/results.json | NVIDIA Ada whitepaper V2.02, Appendix A

MOSS kernel latency

Same unlocked state, identical BF16 inputs, 450 W; 35 samples per implementation; whiskers: observed P10-P90.

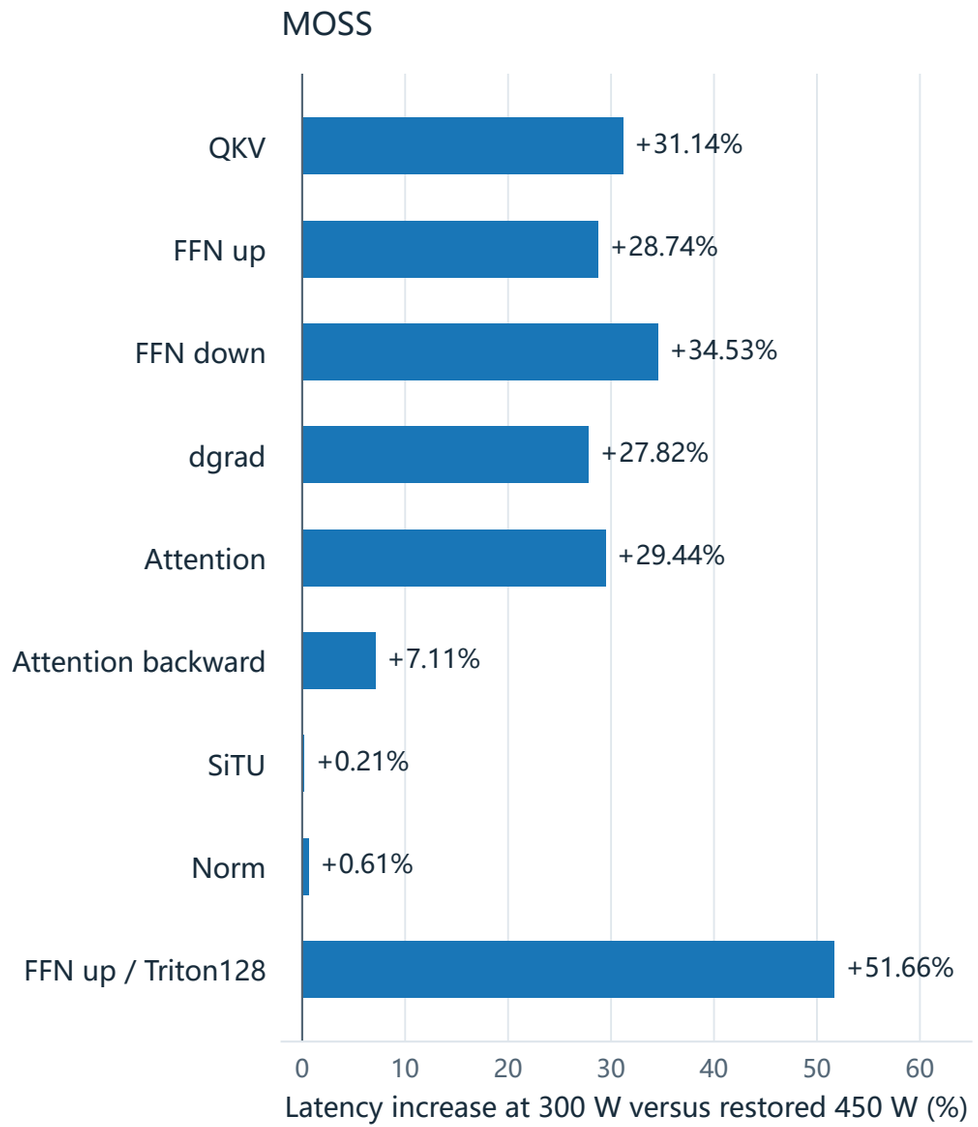


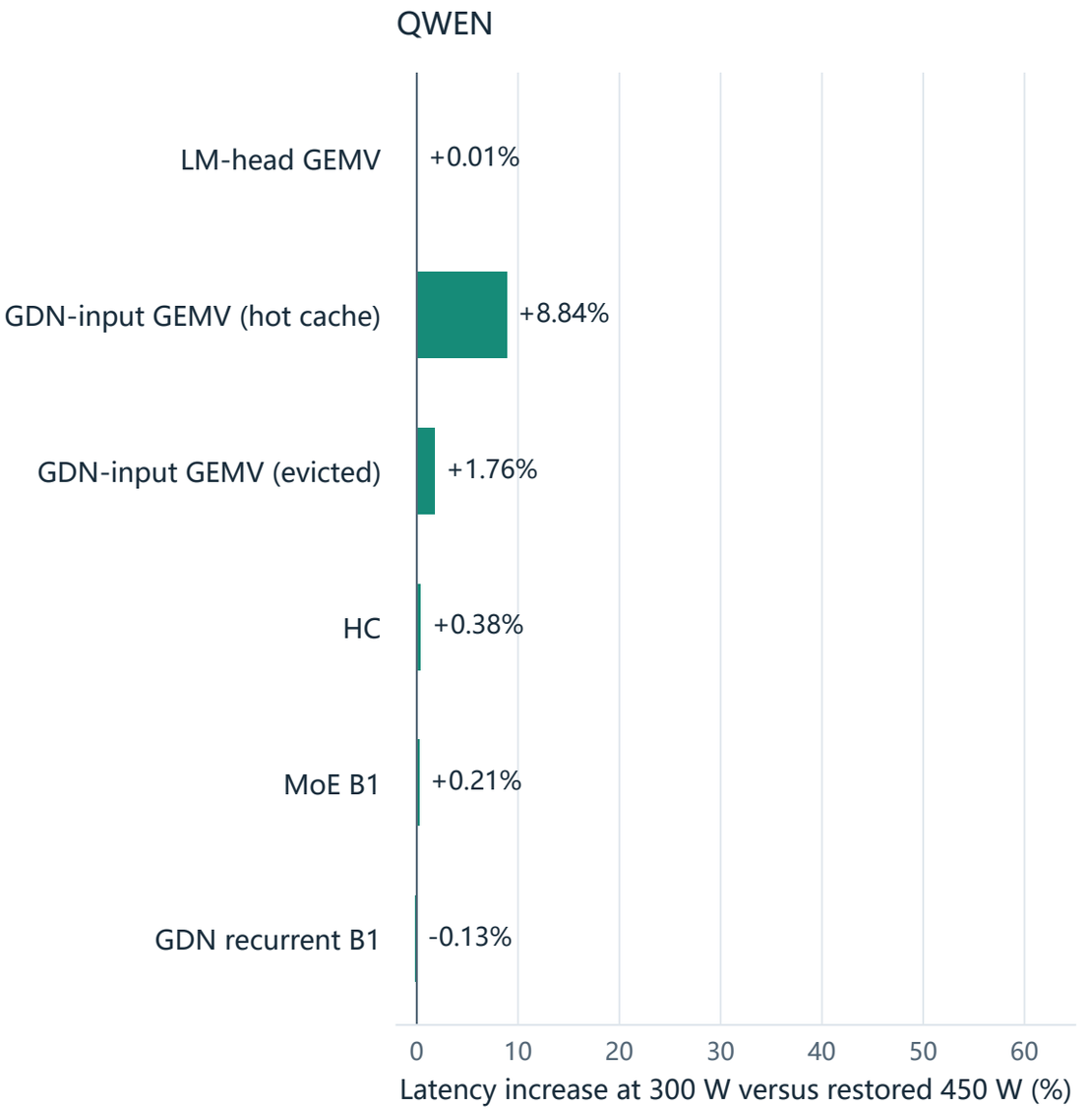
Input shapes are M × N × K.

Source: 2026-10-26/operators.json, paired-g0-450 and paired-g1-450

MOSS / Qwen kernel power sensitivity

Same card and inputs: 300 W versus restored 450 W. Both panels use the same horizontal scale.



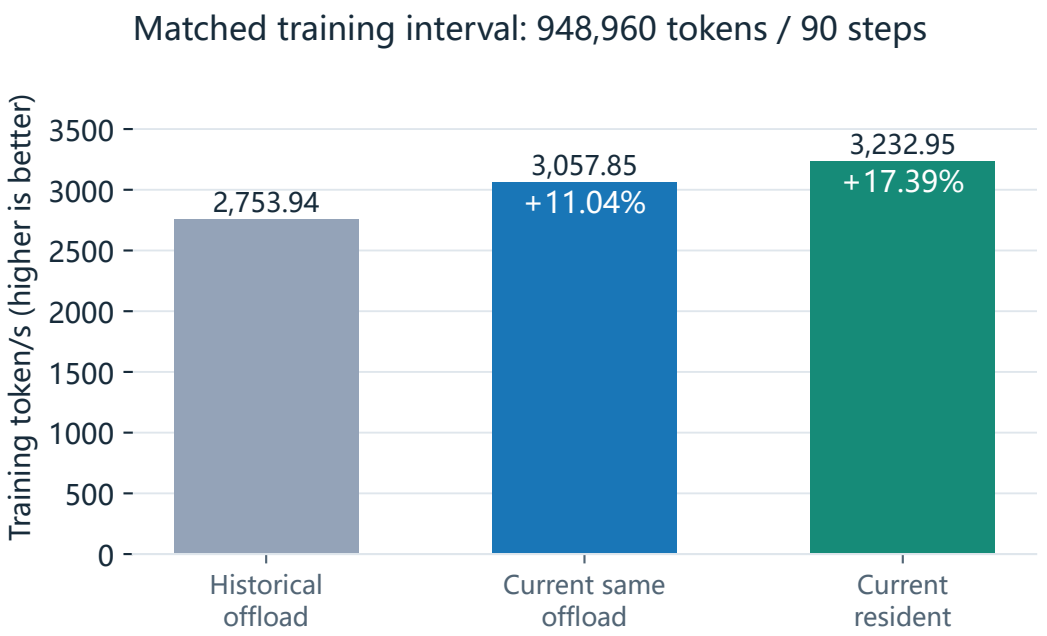


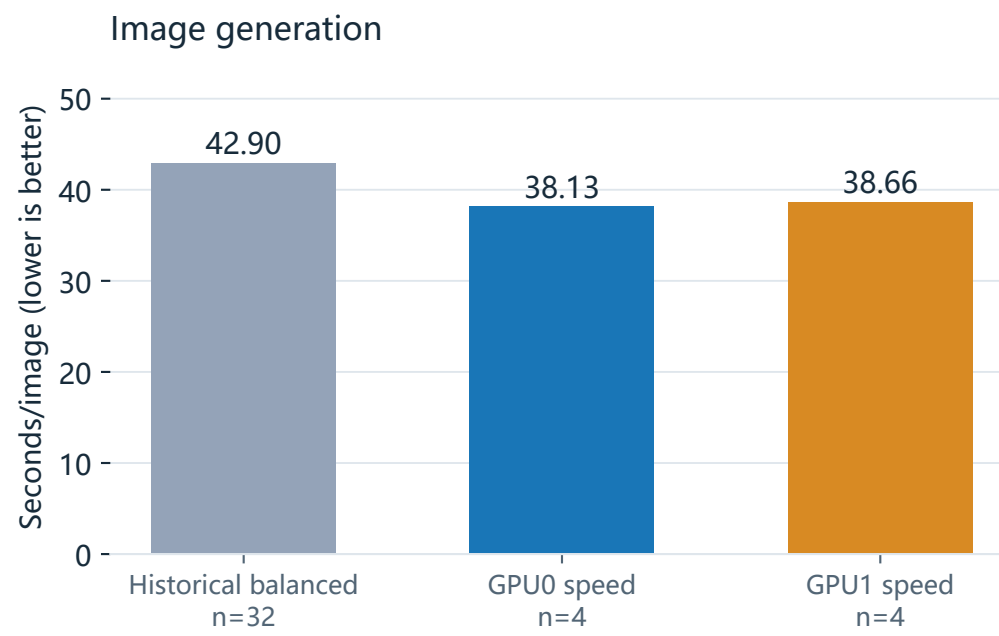
GPU0 · CUDA event timing · ordinary n=21, evicted-cache n=31.

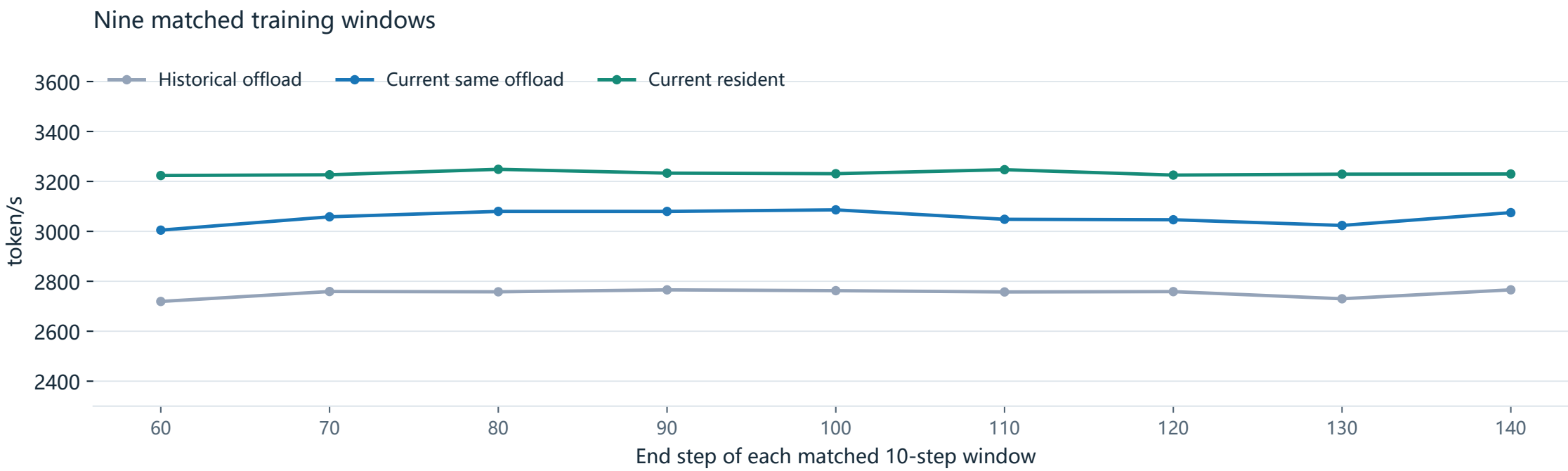
Source: 2026-10-26/operators.json, *-g0-300 and *-g0-450b

MOSS training and inference performance

Training throughput = total tokens / interval time







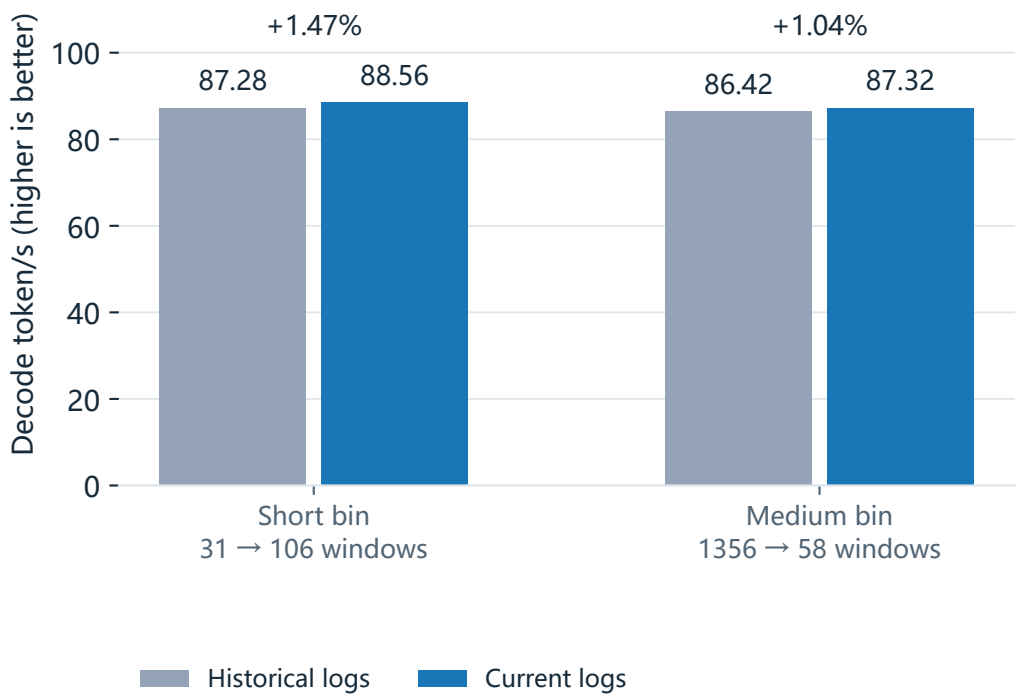
Training interval: steps 51-140 · 90 steps · 237 samples · 948,960 tokens.

Source: 2026-10-26/historical-applications.json

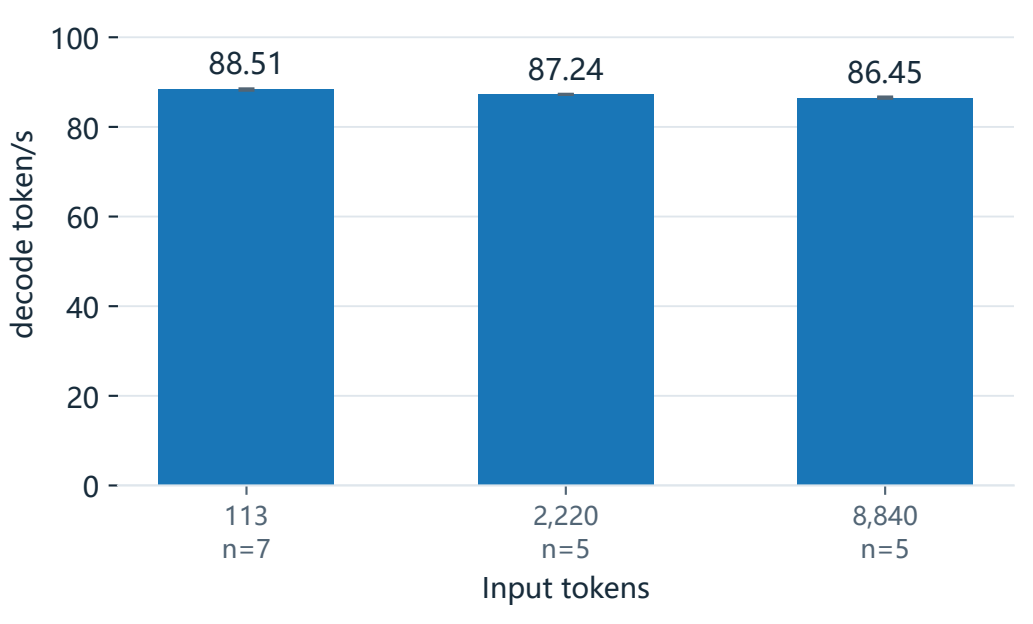
Qwen service performance

C1 · TP2 · prefix-cache hits = 0 · 512 output tokens/request · 2 warm-ups, 17 measured requests

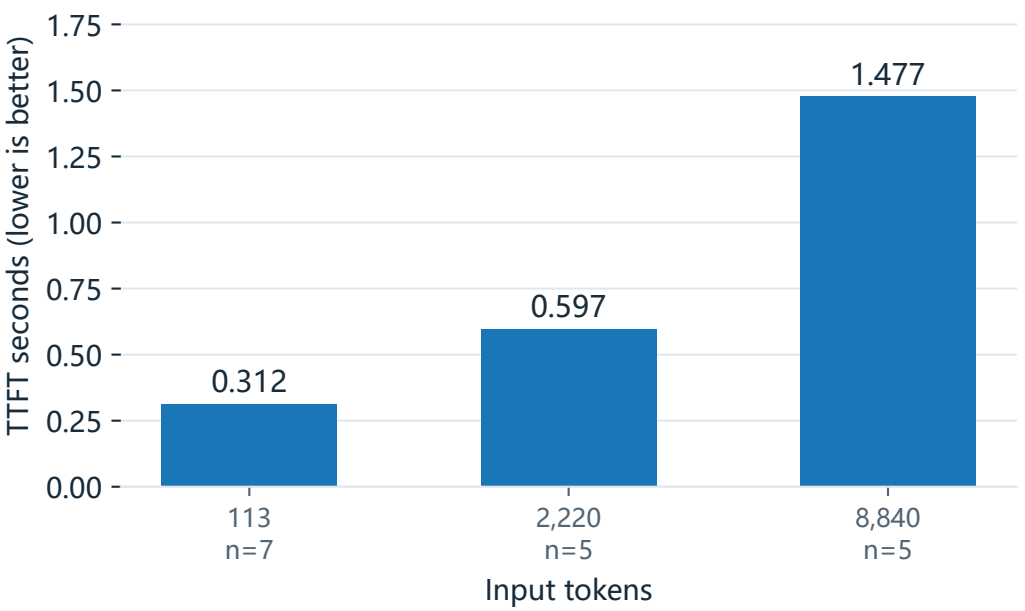
Same-profile service logs



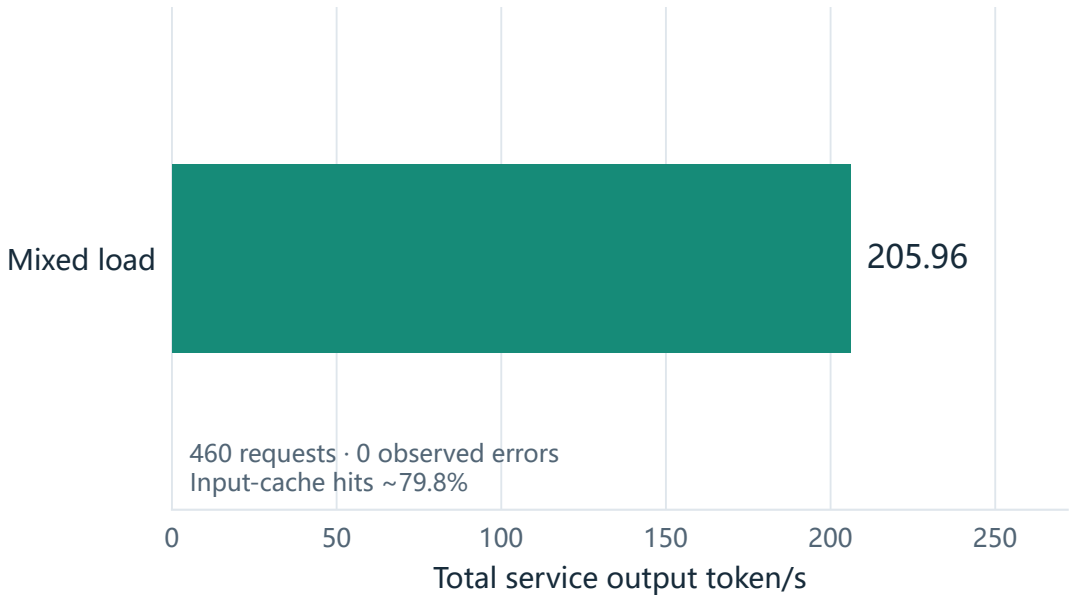
Current concurrency 1: 86-89 token/s



Current concurrency 1: time to first token



Mixed-load total throughput



Current C1 whiskers: observed min-max. Historical log-window counts appear on the x-axis.

Sources: historical-applications.json | qwen-single.json | deployment-summary.json (2026-10-26)